



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Summary of Methodologies

The goal of this SpaceX research was to identify the principal factors in successful rocket landings. The following steps below were used to make this determination:

- 1) **Collected** data using web scraping and the SpaceX REST API
- 2) **Wrangled** data to determine sample counts and create landing outcome variable
- 3) **Explored** data using SQL and Matplotlib visualizations and feature engineer based on analysis
- 4) **Visualized** data using Folium maps and Plotly Dashboards to gain further insights on payload and launch site role in successful landings
- 5) **Created Models** to predict landing outcomes with Logistic Regression, Support Vector Machine, Decision Tree, and K-Nearest Neighbor techniques

Summary of Results

- Launch success rate improved over time
- Launch sites placed close to equator save money on fuel consumption by utilizing Earth rotational speed
- Proximity to coastline and avoidance of cities and major transportation modes help reduce collateral damage cost
- Heavier payload mass correlated with higher success rate
- Some orbit types are easier to achieve success with
- Decision Tree model shows promise with predicting power

Introduction

Background

SpaceX is an American space technology company founded in 2002 by Elon Musk.

Elon's vision was to decrease the cost of space travel in order to make it financially available to the public. They have made several innovations on space technology to make this possible; the most notable of that being the Falcon 9 rocket. What distinguishes this rocket from the rest is its reusability, with the rocket and its 1st stage boosters being able to be recovered and used multiple times. This leads to the average cost per launch for the Falcon 9 hovering around \$67 million, a far cry from the \$150 million of competitors. If the factors that lead to successful landings can be identified and predicted, this would lead to more consistent results and potentially billions of dollars (\$) in cost saved.

Problems to Answer

- Has the success rate of Falcon 9 launches improved over time?
- What factors play a role in determining the success of landings?
- What models can we use to best predict these outcomes?



Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - Call SpaceX REST API and web-scrape Wikipedia tables to collect information on Falcon 9 launch records and put into dataset
- Perform data wrangling
 - Handle missing values, one hot encode categorical variables, create landing outcome variable
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Split the features into train and test data, create 4 models and fit with train data, use models to predict on test data, evaluate performance using confusion matrix + accuracy

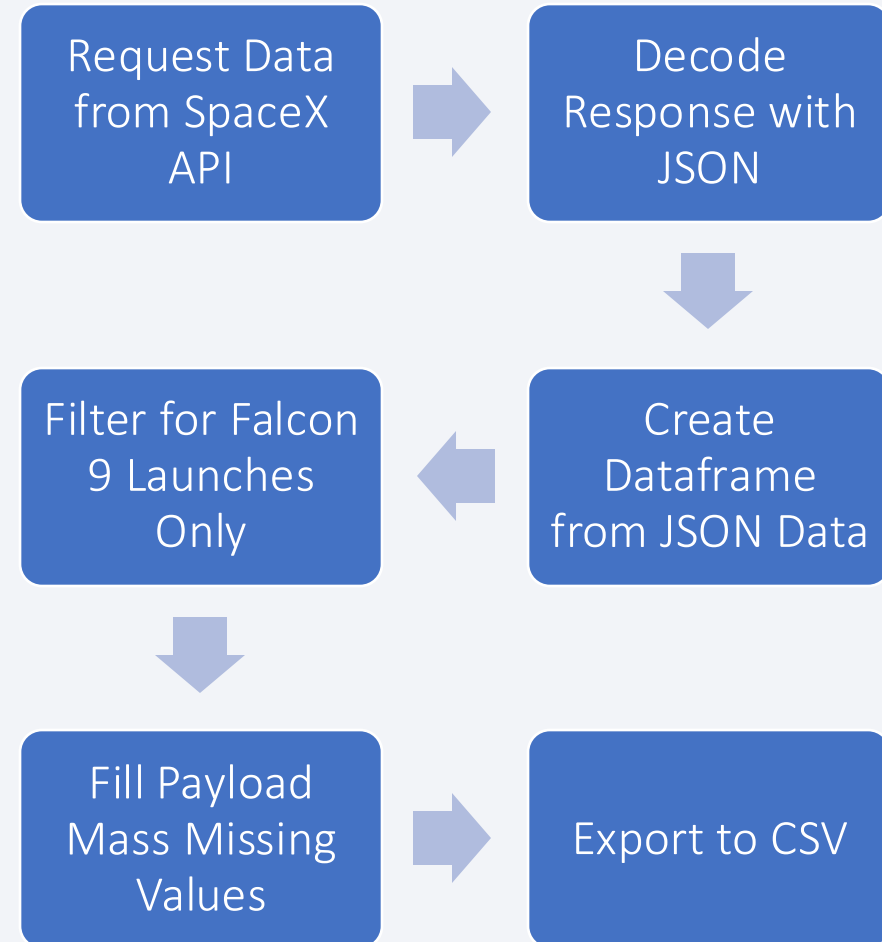


Data Collection

- Data was collected on the SpaceX Falcon 9 launches and landings in two ways:
 - Method 1: SpaceX REST API
 - This takes the JSON response directly from the company application programming interface
 - Method 2: Web scraping
 - This takes the HTML text from the Wikipedia website containing Falcon launch record tables

Data Collection – SpaceX API

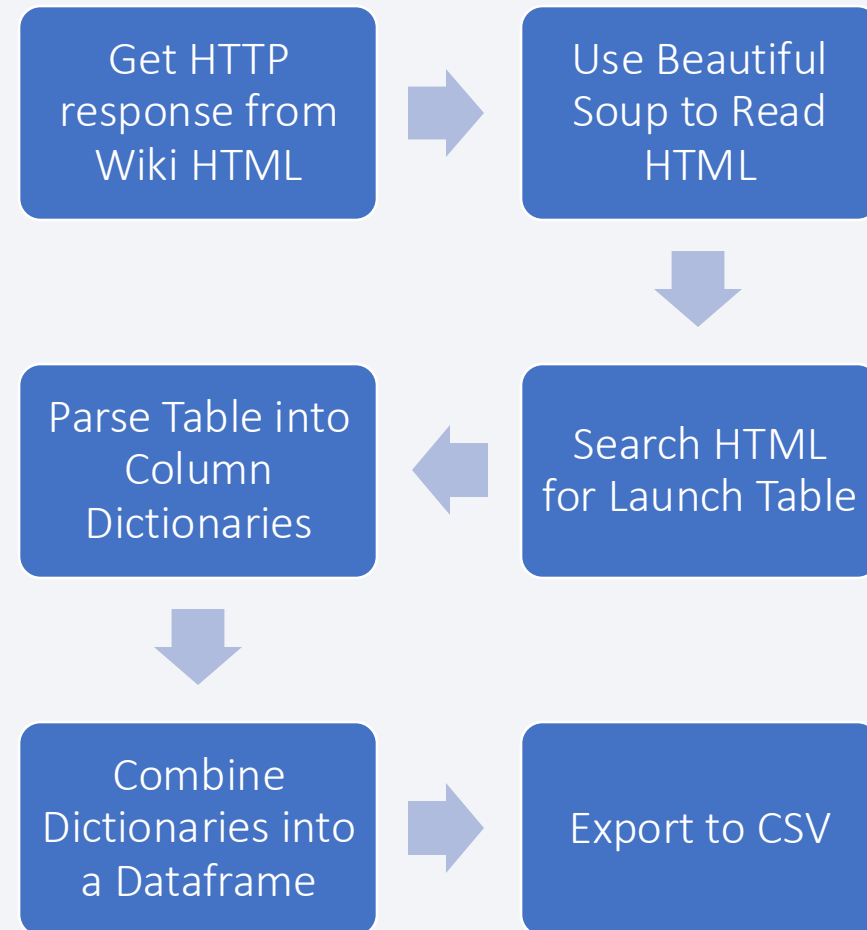
- Flowchart describes process
- Python Libraries Used:
 - Requests
 - NumPy
 - Pandas
 - Datetime



[Data Collection SpaceX API Link](#)

Data Collection - Scraping

- Flow chart describes process
- Python Libraries Used:
 - Requests
 - BeautifulSoup
 - Regex (re)
 - UncodedData
 - Pandas



[Data Collection Web Scraping GitHub Link](#)

Data Wrangling

Process Steps:

Identify null values and distinguish categorical and numerical variables in dataset

Count launches by site, orbit, and landing outcomes

Dummify landing outcome variable using 1 for True (success) and 0 for False (failure) values

Calculate success rate (%) using dummy variable

Export cleaned data to CSV

[Data Wrangling GitHub Link](#)

Landing Outcomes	Success	Failure
Ocean	True Ocean	False Ocean
Ground	True RLTS	False RLTS
Drone	True ASDS	False ASDS

Table: Mission outcome variables in initial dataset explained

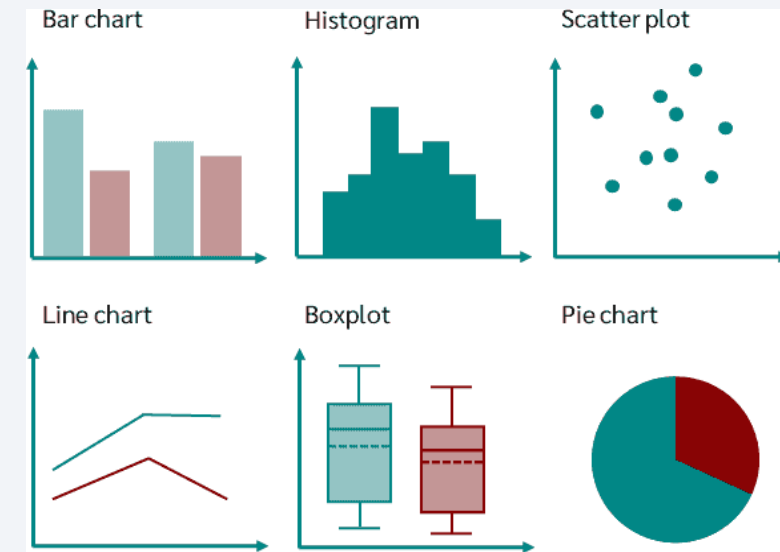
EDA with Data Visualization

Reasoning:

- Use scatter plots to identify relationships, if any, that exist among the variables in order to guide directions for machine learning modeling
- Use bar charts to identify correlation between categorical variables and landing success
- Line chart to see how variables change over time

Scatter Plots used in exploratory analysis:

- Flight Number vs Launch Site
- Payload vs Launch Site
- Flight Number vs Orbit Type
- Payload vs Orbit Type



Other charts used:

- **Line chart** measuring success rate over time
- **Bar chart** identifying success rate relative to orbit type

[EDA Visualization GitHub Link](#)

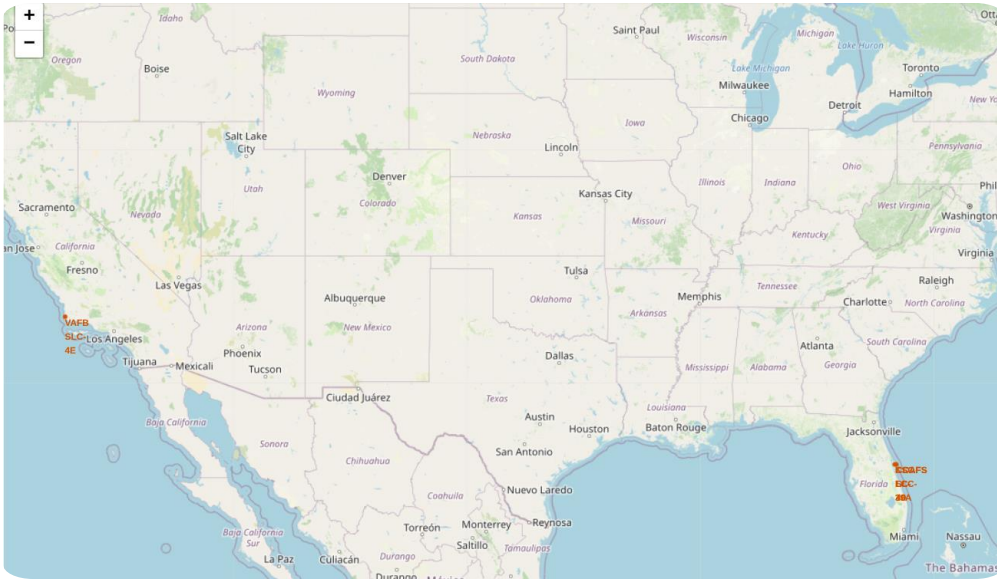


EDA with SQL Querying

- Unique launch site names
- 5 Records from launch sites beginning with "CCA"
- Payload mass of NASA-launched boosters
- Booster F9 version 1.1 average payload mass
- First successful ground pad landing date
- Names of boosters between $4000 \text{ kg} < \text{payload mass} < 6000 \text{ kg}$ and successfully landed by drone ship
- Total number of failed and successful missions
- Names of booster versions that carried max payload
- Records of failed drone landings in months of 2015
- Ranked landing outcome list between 06/04/2010 and 03/20/2017

• [EDA SQL Queries GitHub Link](#)

Interactive Map with Folium



- **Orange circles** placed on all launch site areas using its name and coordinates
- Colored info marker clusters added to each site with:
 - **Green = Successful mission**
 - **Red = Failed mission**
- Distance lines in KM added to show **CCAFS LC-40** distances from nearest **city, railway, highway,** and **coastline**

[Interactive Visualization Folium GitHub Link](#)

Dashboard with Plotly Dash

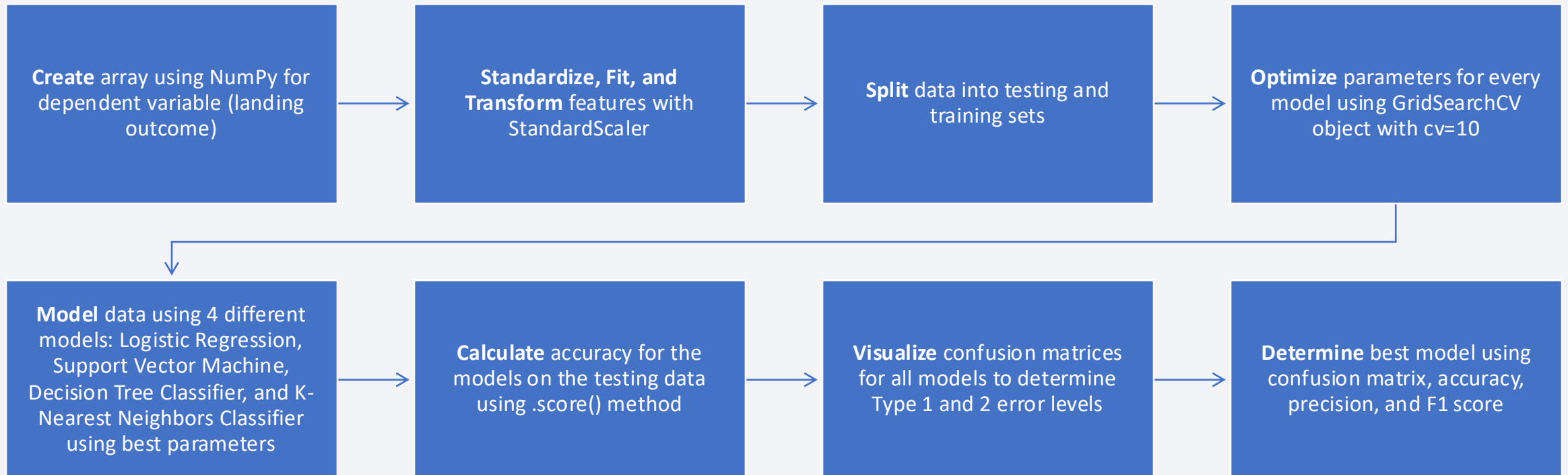
Dashboard Features:

- **Launch Site Drop Down Selector**
 - Graphs for all sites or selected site
- **Successful Launch Pie Charts**
 - Displays success/fail ratios for individually selected launch site
- **Payload Mass vs Success Rate Scatter Plot**
 - Colored by Booster Version
 - **Payload Mass Adjustable Slider**

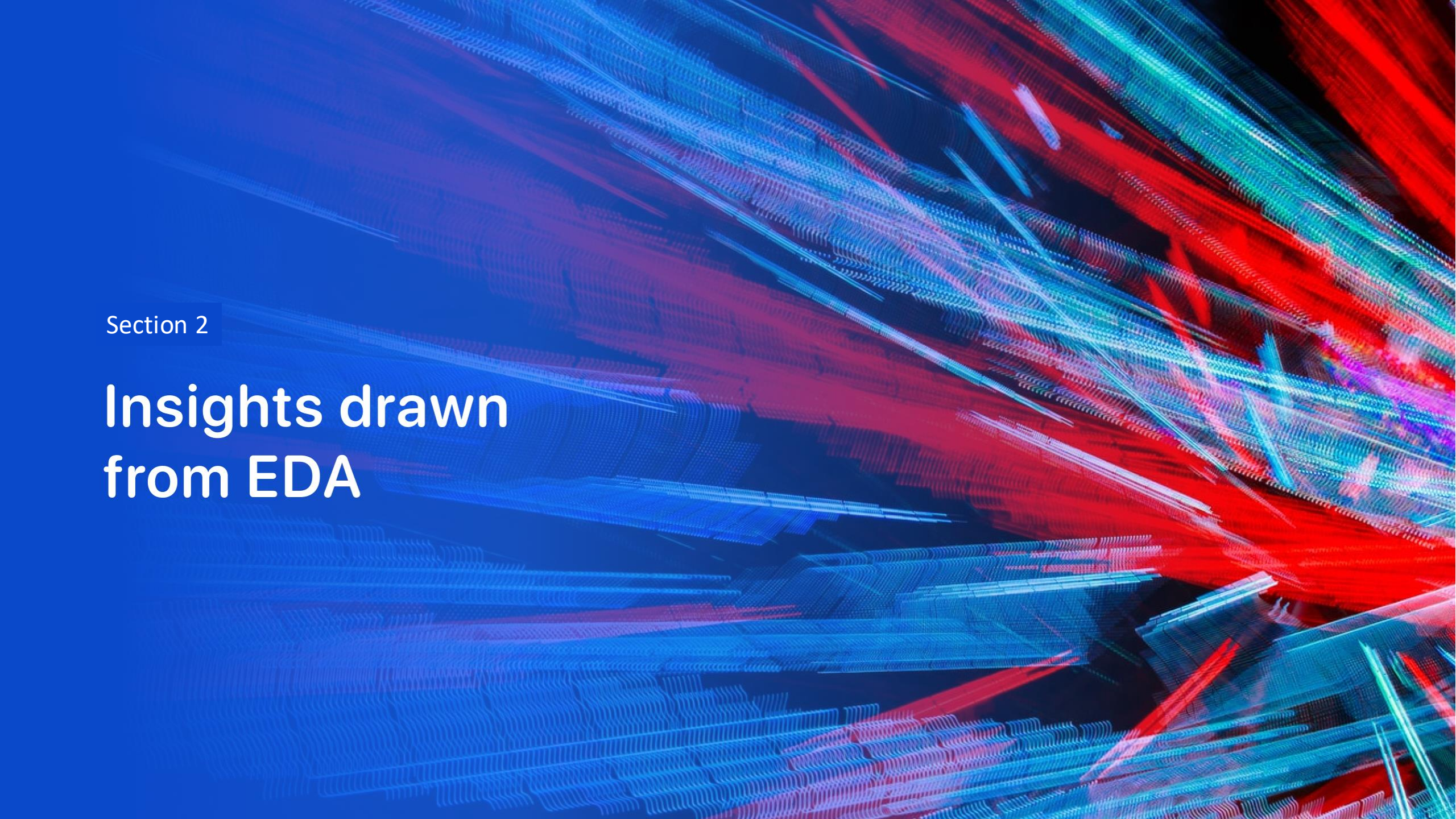
[Data Visualization Plotly Dashboard GitHub Link](#)



Predictive Analysis (Classification)



[Machine Learning Prediction GitHub Link](#)

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of blue and red, creating a sense of motion or data flow. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is high-tech and digital.

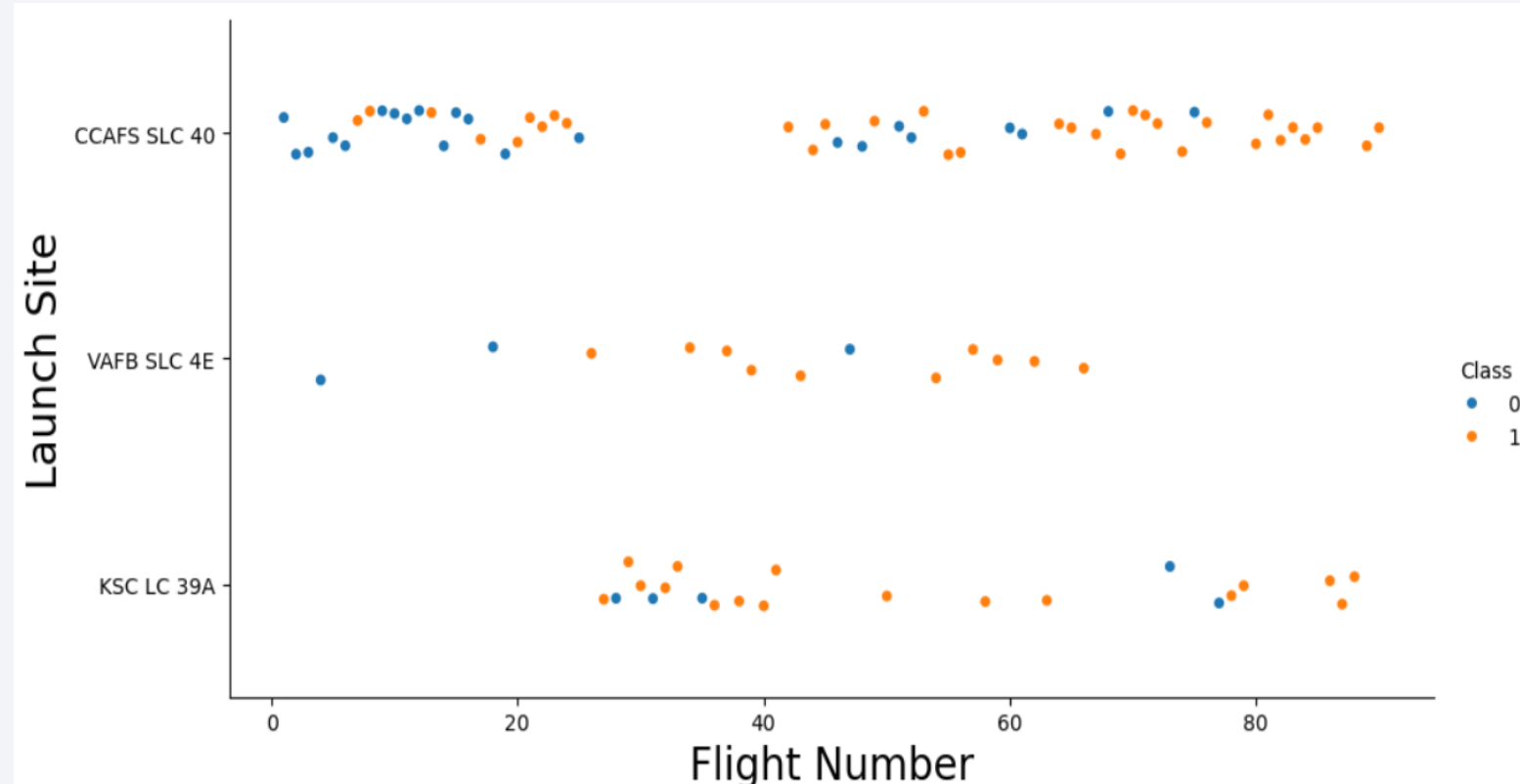
Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

Analysis

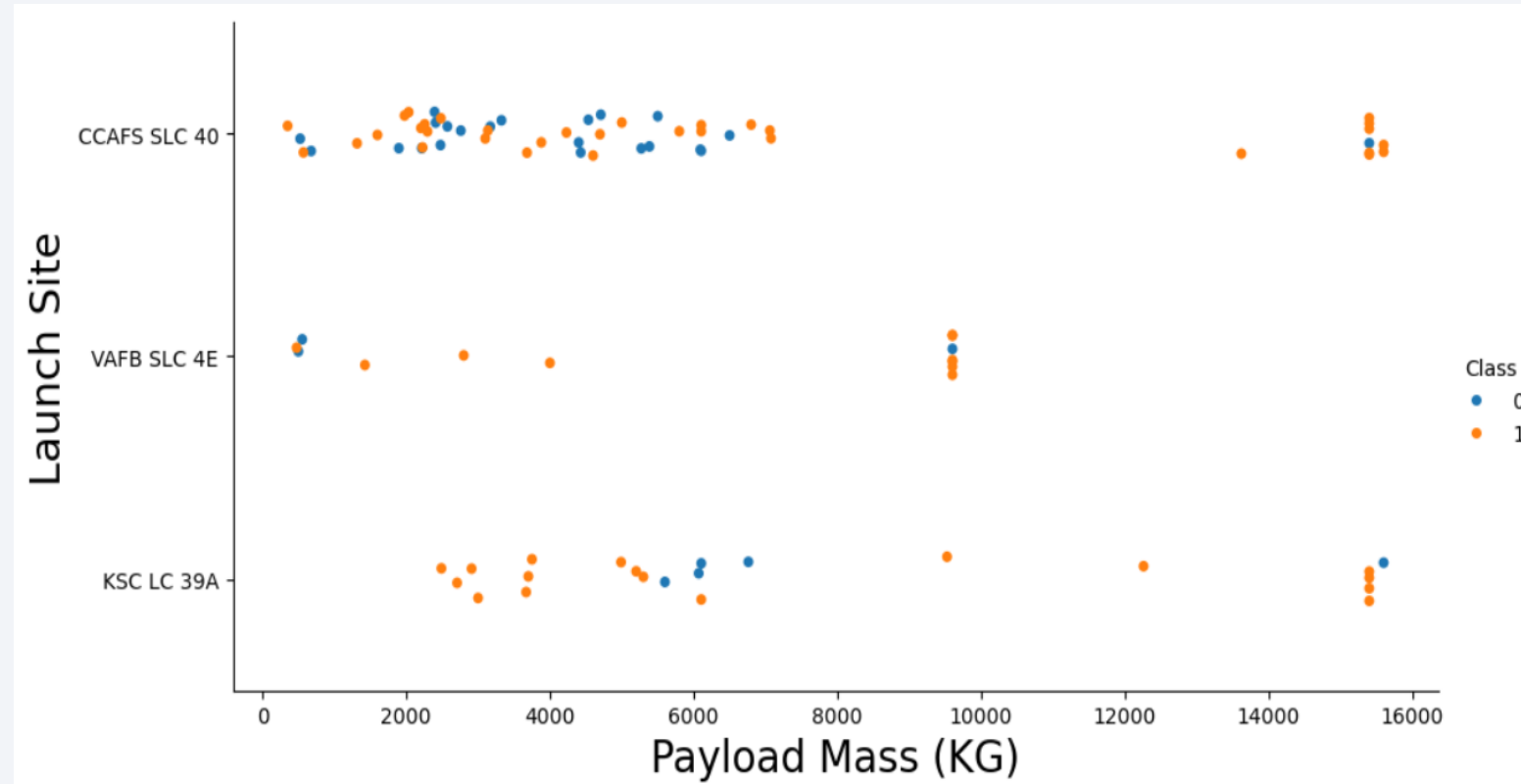
- **Blue = Failed** **Orange = Successful**
- ~50% of launches and 86% of initial 25 launches took place at CCAFS
- VAFB and KSC had higher success rate
- Later launch attempts appeared to be more successful than earlier ones across all 3 launch sites



Payload vs. Launch Site

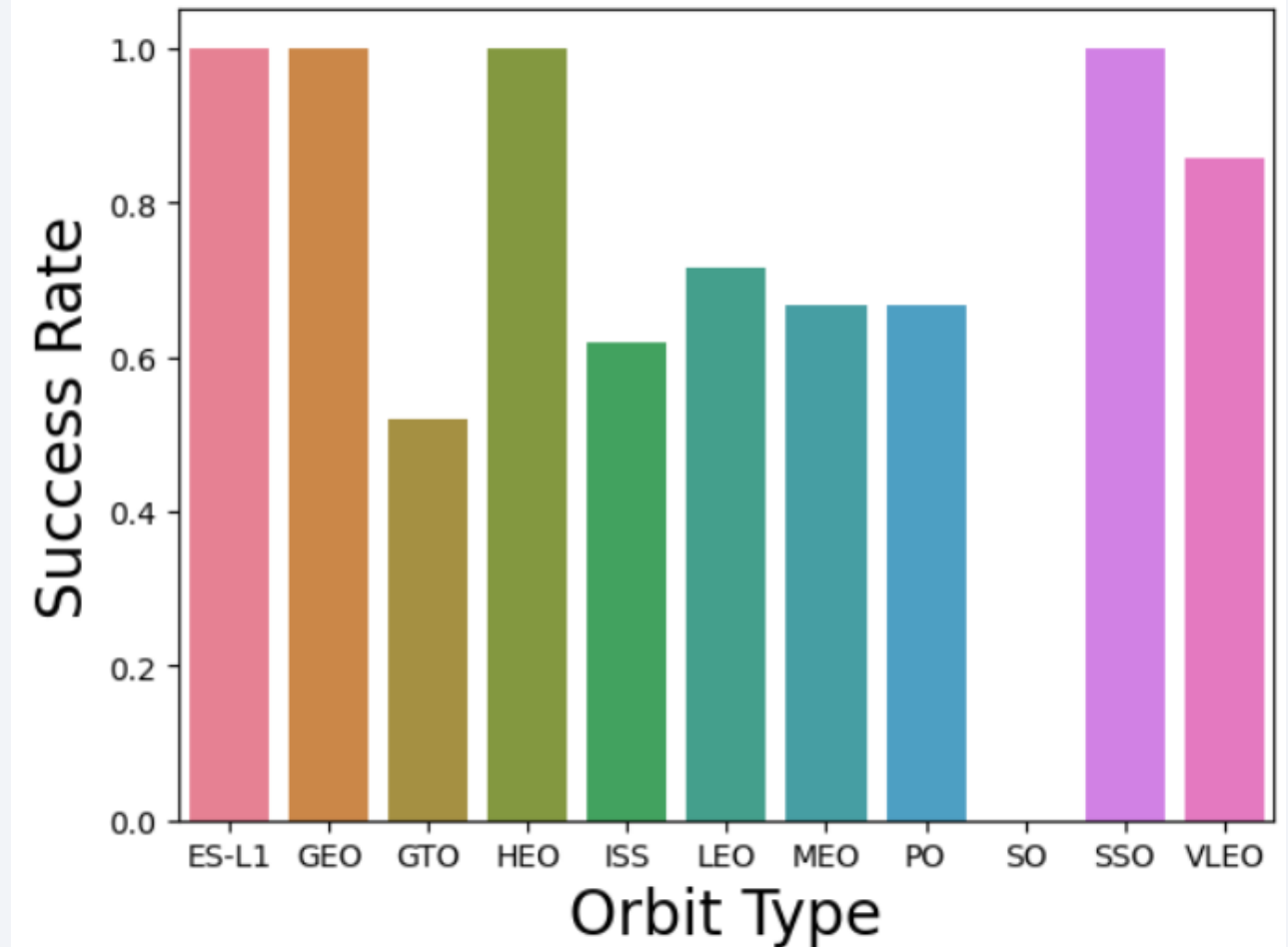
Analysis

- **Blue = Failed** **Orange = Successful**
- No clear relationship between payload mass and success rate
- VAFB launched no rockets with payload greater than 10,000Kg
- High success rate for rockets with >8000Kg payload mass
- All KSC rockets less than ~5500Kg successfully landed



Success Rate vs. Orbit Type

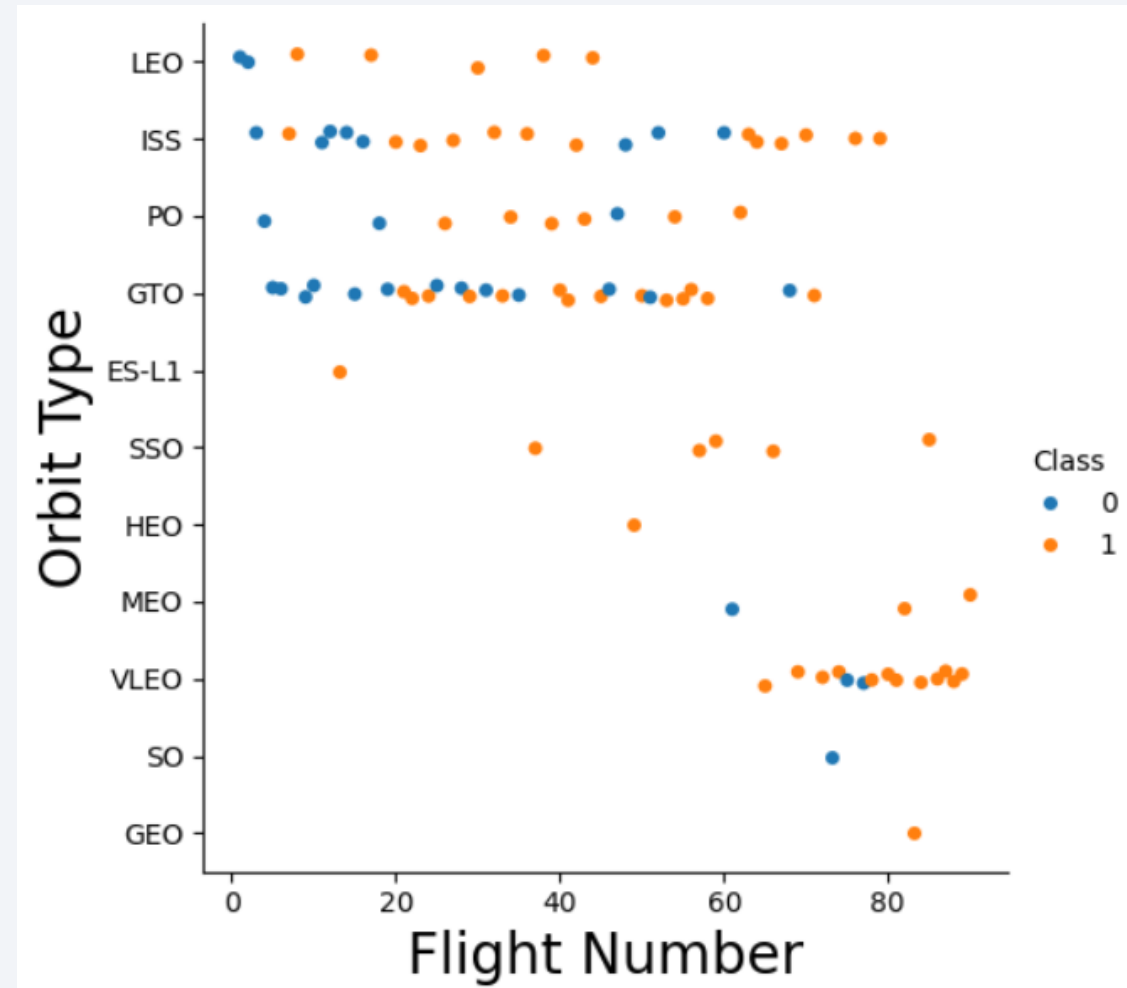
- **0% Success Rate: SO**
- **50-85% Success Rate: GTO, ISS, LEO, MEO, PO, VLEO**
- **100% Success Rate: ES-L1, GEO, HEO, SSO**



Flight Number vs. Orbit Type

Analysis

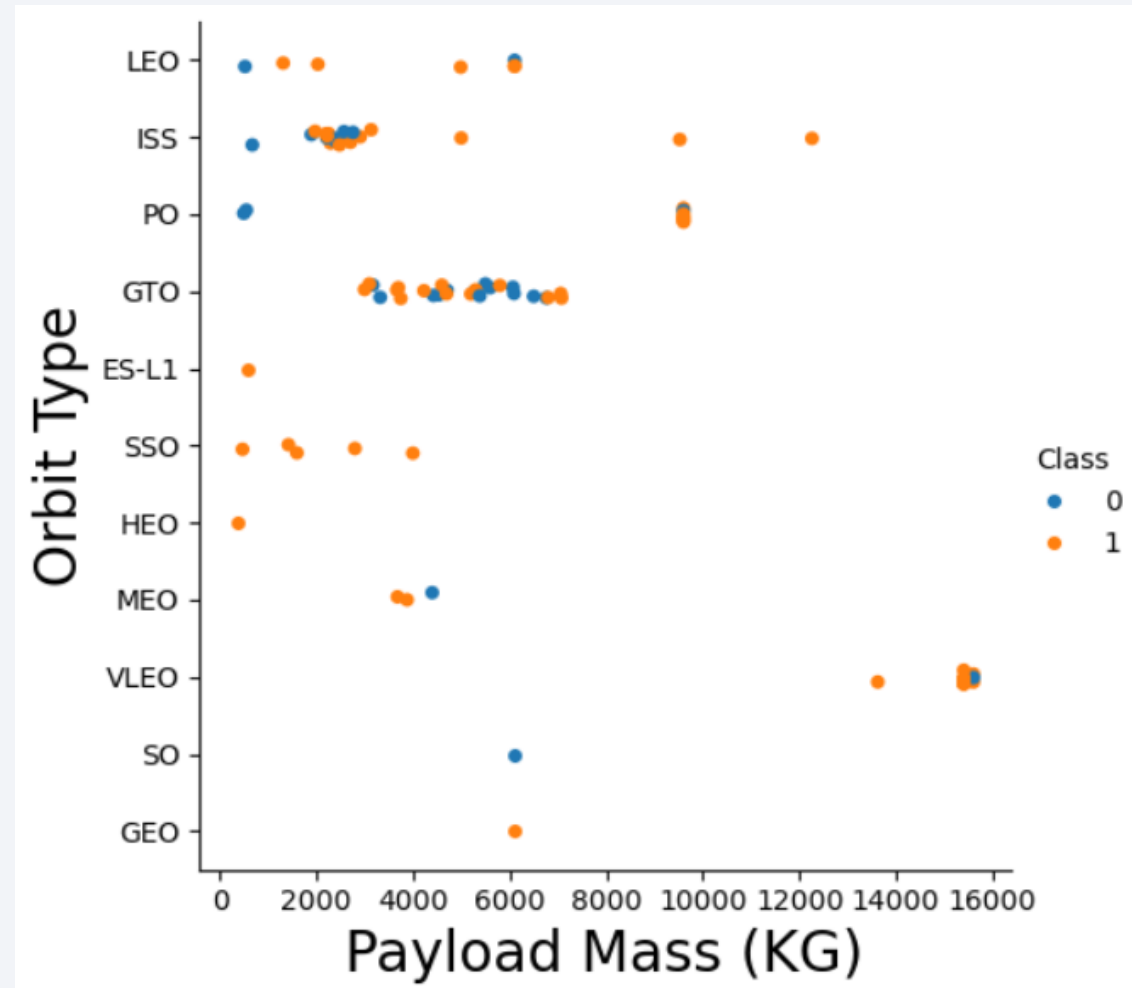
- **Blue = Failed** **Orange = Successful**
- ES-LI1, HEO, SO, and GEO only 1 launch each, $\frac{3}{4}$ successful
- Most early launches were LEO, ISS, PO, and GTO and were unsuccessful
- Most later launches were VLEO and successful
- All 5 SSO flights were successful

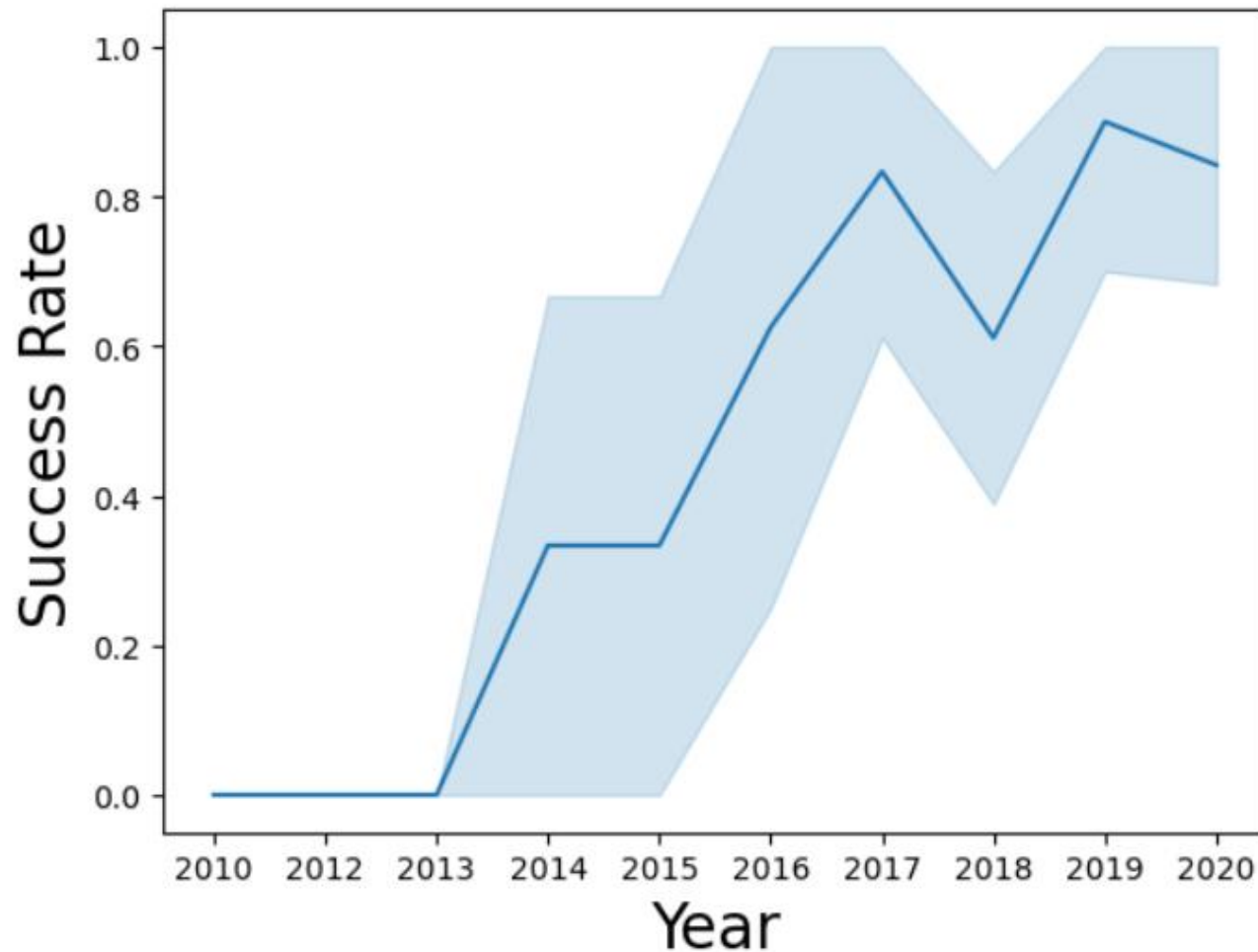


Payload vs. Orbit Type

Analysis

- **Blue = Failed** **Orange = Successful**
- Most launches were below 8000Kg
- GTO launches between 3000Kg and 8000Kg and had mixed success
- VLEO launched exclusively with payloads > 13,000Kg, only orbit type to have that heavy of payloads, mostly successful
- SSO all successful and payload < 5,000Kg
- ISS, LEO, and PO had more success on average with increasing payload mass





Launch Success Yearly Trend

- Launch success drastically improved over time
- 2010-2013 saw no successful launches
- 2016-2017 year range that showed highest improvement range
- Peak success rate: ~90% in 2019

Launch Site Information

```
%sql select distinct Launch_Site from SPACEXTBL
```

```
* sqlite:///my_data1.db  
Done.
```

Launch_Site

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

Launch Site List

- CCAFS LC-40
- VAFB SLC-4E
- KSC LC-39A
- CCAFS SLC-40

First 5 launch records from CCAFS LC-40 or CCAFS SLC-40

```
%sql select * from SPACEXTBL where Launch_Site like 'CCA%' limit 5
```

```
* sqlite:///my_data1.db  
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Payload Mass

Total NASA (CRS) Payload Mass = 45596KG

Average F9v1.1 Booster Payload Mass = 2928.4KG

```
%sql select Customer, sum(PAYLOAD_MASS_KG_) from SPACEXTBL where Customer == 'NASA (CRS)'
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Customer	sum(PAYLOAD_MASS_KG_)
NASA (CRS)	45596

```
%sql select Booster_Version, avg(PAYLOAD_MASS_KG_) from SPACEXTBL where Booster_Version == 'F9 v1.1'
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Booster_Version	avg(PAYLOAD_MASS_KG_)
F9 v1.1	2928.4

First Successful Ground Landing Date

Earliest Ground Pad Landing Success Date: 12-22-2015

```
%%sql select min(Date) as first_record, * from SPACEXTBL
where Landing_outcome == 'Success (ground pad)'
```

```
* sqlite:///my_data1.db
```

Done.

first_record	Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2015-12-22	2015-12-22	1:29:00	F9 FT B1019	CCAFS LC-40	OG2 Mission 2 11 Orbcomm-OG2 satellites	2034	LEO	Orbcomm	Success	Success (ground pad)

Successful Drone Ship Landing with Payload between 4000 and 6000

Payloads that matched criteria:

- JCSAT-14
- JCSAT-16
- SES-10
- SES-11 / EchoStar105

```
%%sql select Payload from SPACEXTBL
      where Landing_Outcome = 'Success (drone ship)' and PAYLOAD_MASS__KG_ between 4000 and 6000;

* sqlite:///my_data1.db
Done.
```

Payload
JCSAT-14
JCSAT-16
SES-10
SES-11 / EchoStar 105

```
%sql select Mission_Outcome, count(Mission_Outcome) as Frequency from SPACEXTBL group by Mission_Outcome
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Mission_Outcome	Frequency
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Total Number of Successful and Failure Mission Outcomes

Total Success Counts:

- 99 Successes
- 1 Success with payload status unclear
- 1 Failure

Boosters Carried Maximum Payload

```
%%sql select Booster_Version from SPACEXTBL
      where PAYLOAD_MASS__KG_ = (select max(PAYLOAD_MASS__KG_) from SPACEXTBL) order by Booster_Version
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Booster_Version

F9 B5 B1048.4

F9 B5 B1048.5

F9 B5 B1049.4

F9 B5 B1049.5

F9 B5 B1049.7

F9 B5 B1051.3

F9 B5 B1051.4

F9 B5 B1051.6

F9 B5 B1056.4

F9 B5 B1058.3

F9 B5 B1060.2

F9 B5 B1060.3

Max Payload Boosters:

- **F9 B5**
 - **B1048.4**
 - **B1048.5**
 - **B1049.4**
 - **B1049.5**
 - **B1049.7**
 - **B1051.3**
 - **B1051.4**
 - **B1051.6**
 - **B1056.4**
 - **B1058.3**
 - **B1060.2**
 - **B1060.3**

2015 Launch Records

All Drone Ship Landing Failures in 2015

```
%%sql select substr(Date,0,5) as Year, substr(Date,6,2) as Month, Landing_Outcome, Booster_Version, Launch_Site from SPACEXTBL
      where substr(Date,0,5) == '2015' and Landing_Outcome == 'Failure (drone ship)'
```

```
* sqlite:///my_data1.db
```

Done.

Year	Month	Landing_Outcome	Booster_Version	Launch_Site
2015	01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
2015	04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Types of landing outcomes, ranked by frequency, occurring between June 4th, 2010 and March 20, 2017

```
%%sql select Landing_Outcome, count(*) as Frequency from SPACEXTBL
      where Date between '2010-06-04' and '2017-03-20' group by Landing_Outcome order by Frequency desc
```

```
* sqlite:///my_data1.db
```

Done.

Landing_Outcome	Frequency
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

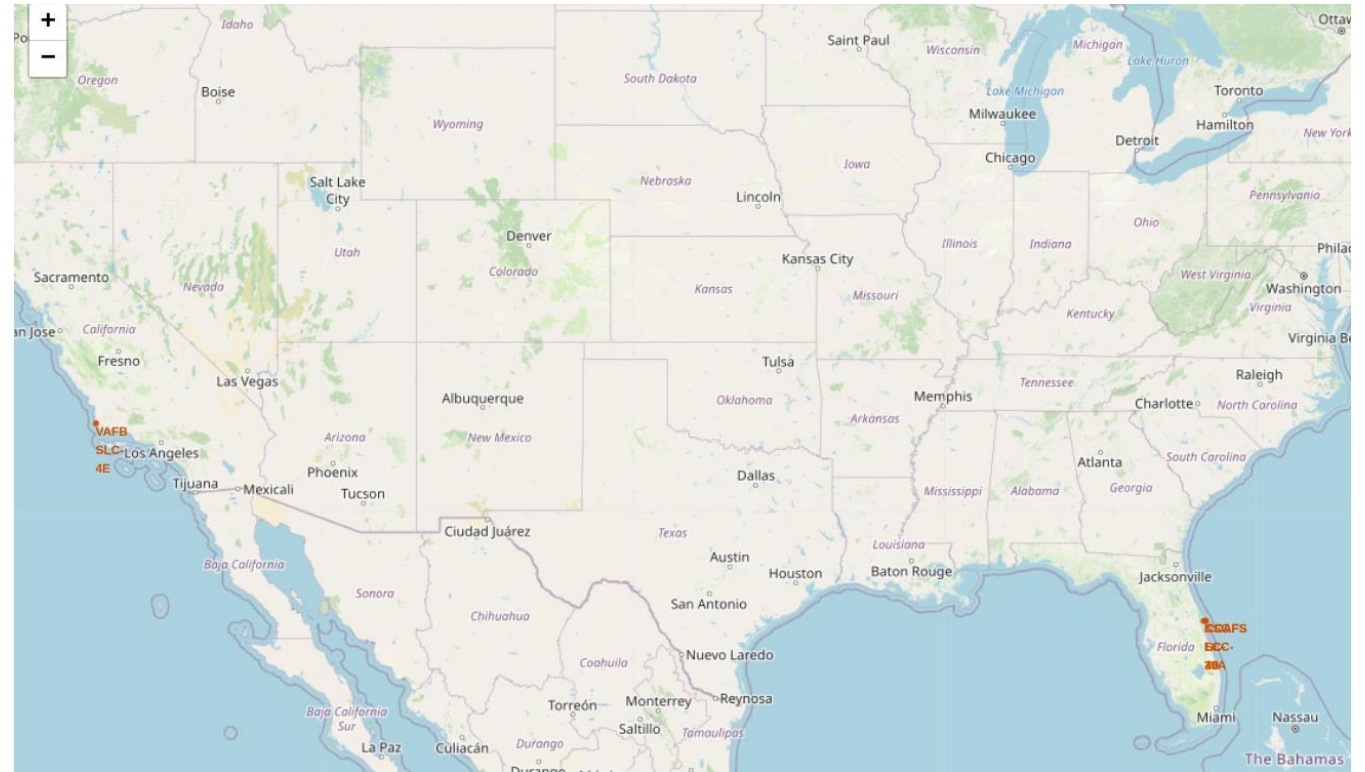
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue space with stars. The Earth's surface is dark blue, with bright yellow and orange lights from cities and towns. The lights are concentrated in the lower right quadrant of the image, following the curve of the Earth.

Section 3

Launch Sites Proximities Analysis

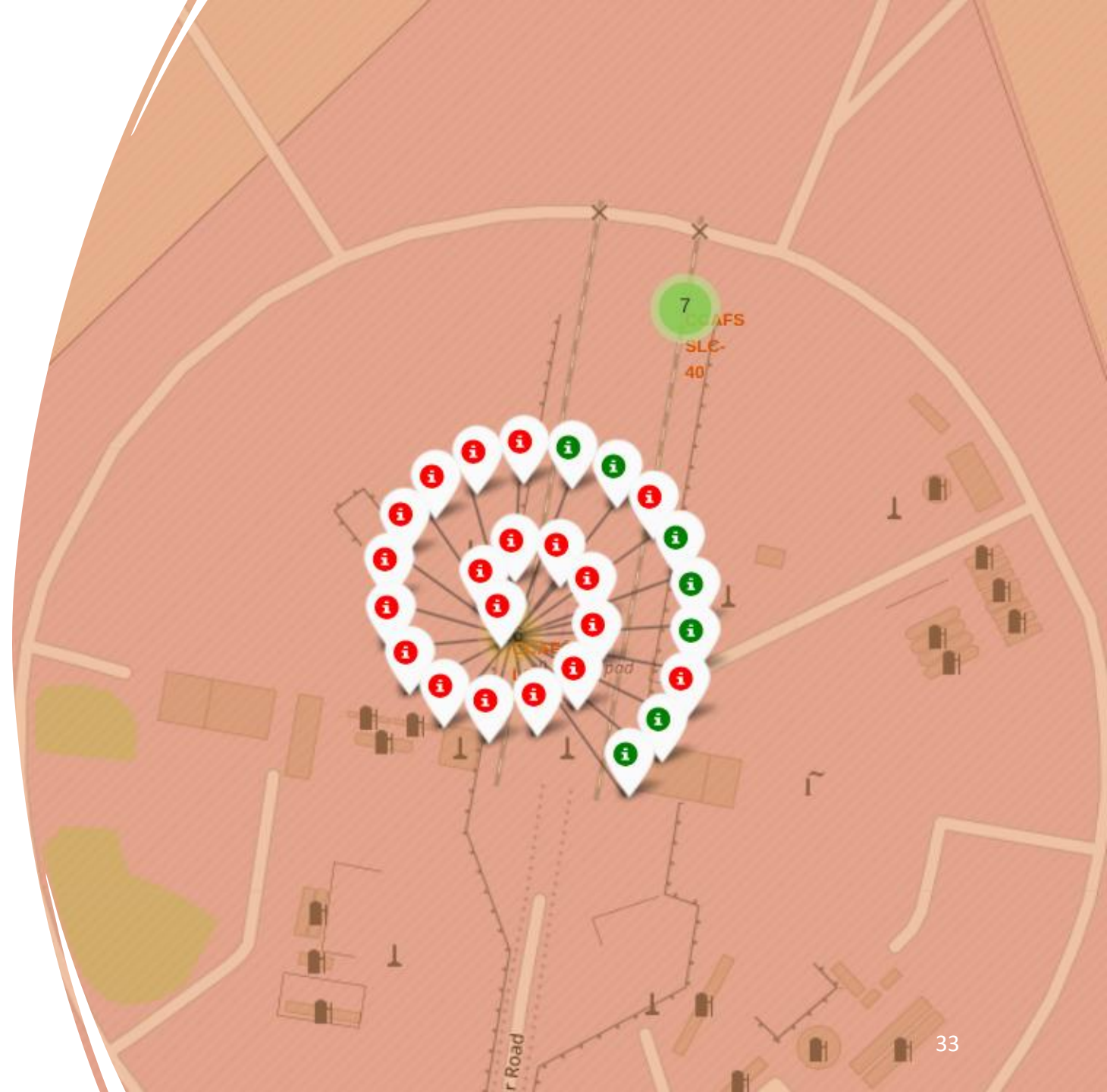
Launch Site Locations Around the US

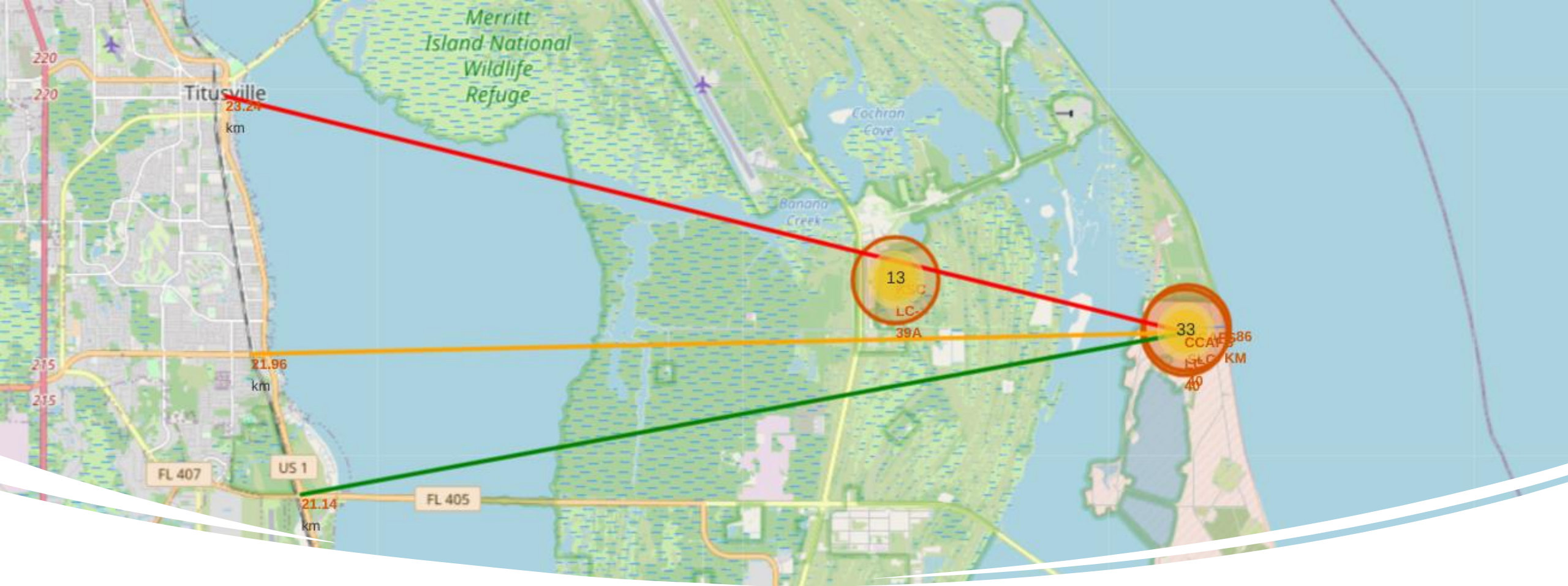
- **Orange Markers indicate launch sites**
- **Built near coastline and away from major metropolitans**, likely to avoid collateral damage
- **Southern most regions of US**, Earth rotational speed is highest near Equator, so placing launch sites closer to Equator within US soil can help minimize fuel consumption



Launch Outcomes at Each Site

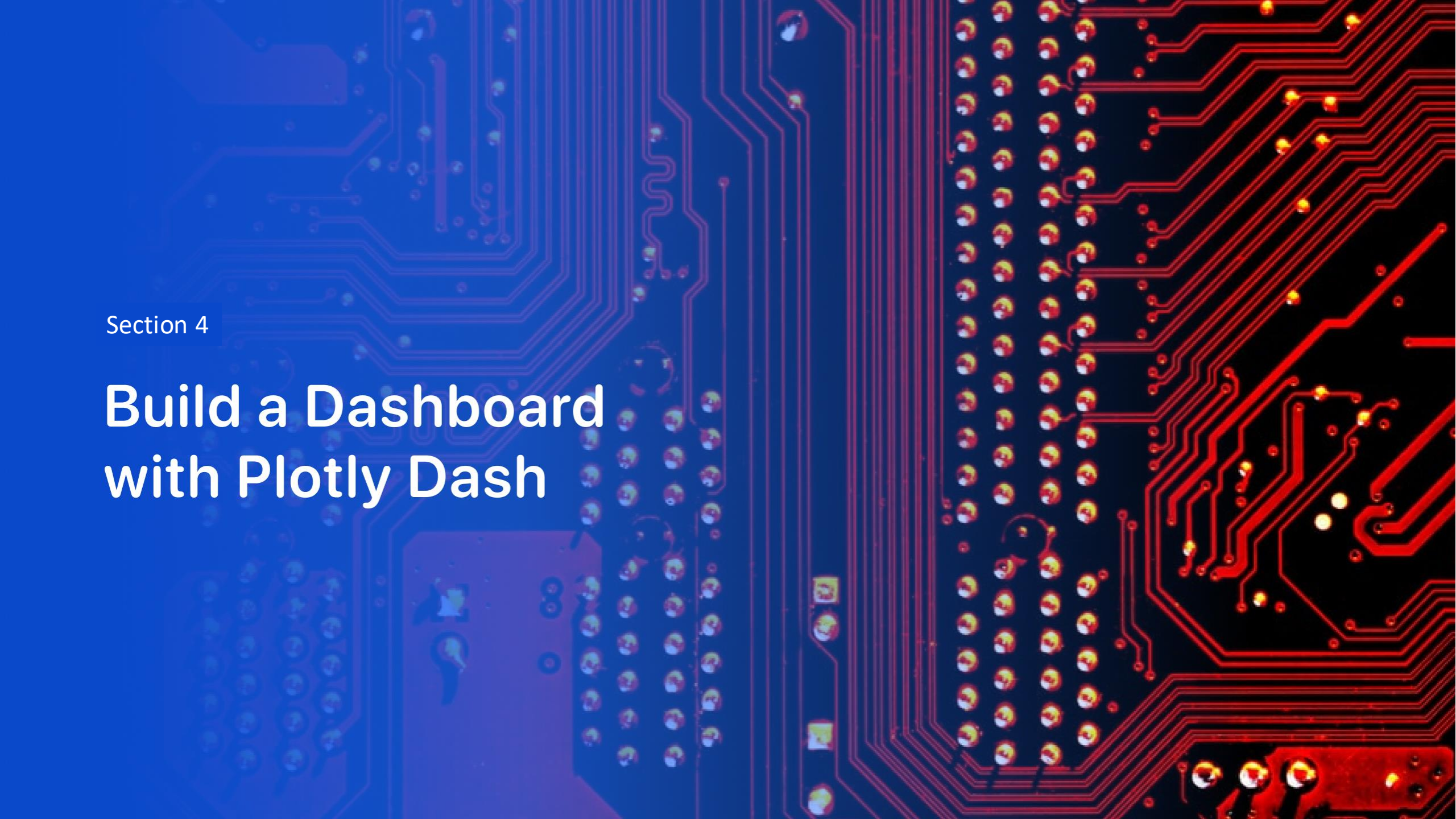
- Info markers indicate site outcomes
- **Green = Success** **Red = Failure**
- Success rates by site:
 - CCAFS LC-40 = 26.92% (3/7)
 - CCAFS SLC-40 = 42.85% (7/26)
 - KSC LC-39A = 76.92% (10/13)
 - VAFB SLC-4E = 40.00% (4/10)





Proximities to Transportation

- CCAFS LC-40 proximities shown
- **Nearest City: Titusville - 23.24KM**
- **Nearest Railway: Florida East Coast Railway - 21.96 KM**
- **Nearest Highway: US 1 - 21.14 KM**
- **Nearest Coastline – 0.86 KM**
- Close to ocean, equally far from cities, railways, and highways



Section 4

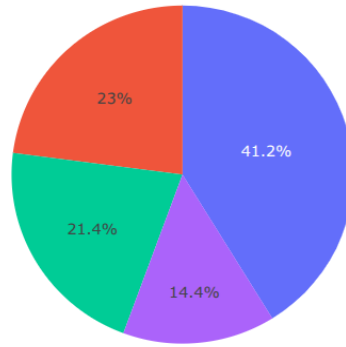
Build a Dashboard with Plotly Dash

SpaceX Launch Records Dashboard

All Sites

×

Total Success Launches by Site



- KSC LC-39A
- CCAFS SLC-40
- VAFB SLC-4E
- CCAFS LC-40

Successful Landings By Launch Site

- **KSC LC-39A** has the most successful launches of all 4 sites
- **Uneven distribution** in launch site mission successes suggest differences in rocket design processes and mission orbit types

Success/Fail Ratio for Missions from KSC LC-39A

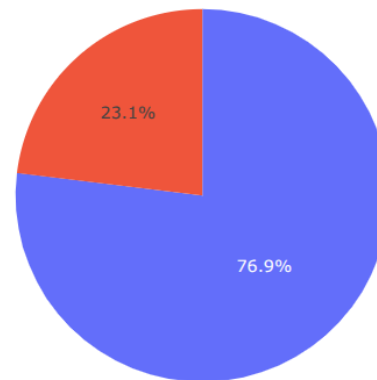
- **KSC LC-39A** had highest success ratio, with **76.92%**
- Of 13 launches, **10 were successful**

SpaceX Launch Records Dashboard

KSC LC-39A

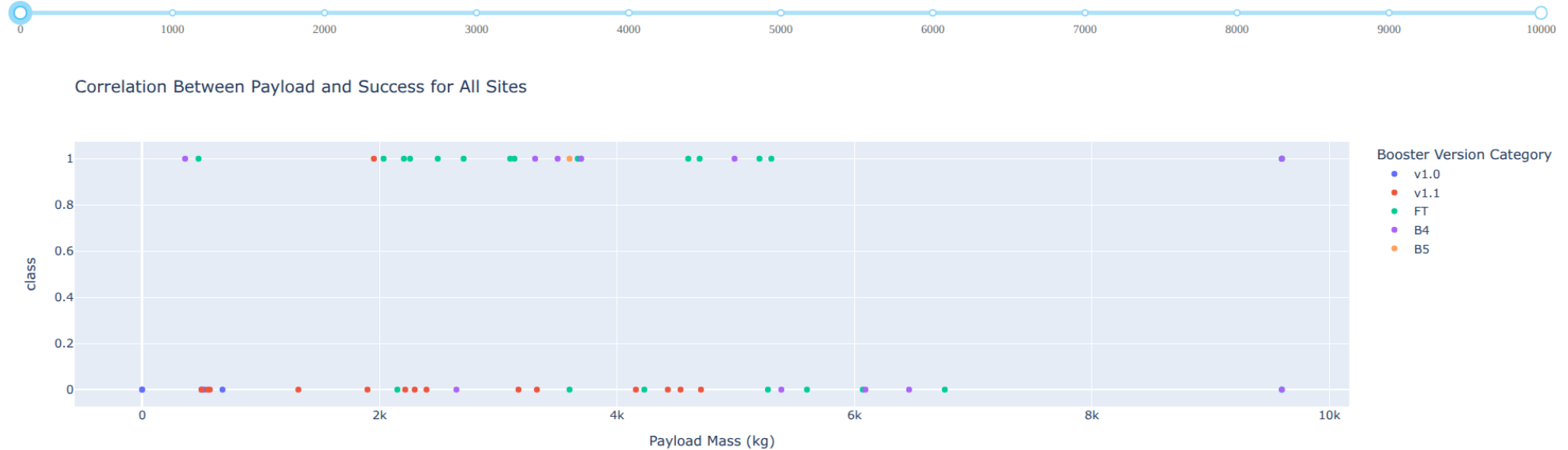
×

Total Success Launches for Site KSC LC-39A



0
1

Payload range (Kg):

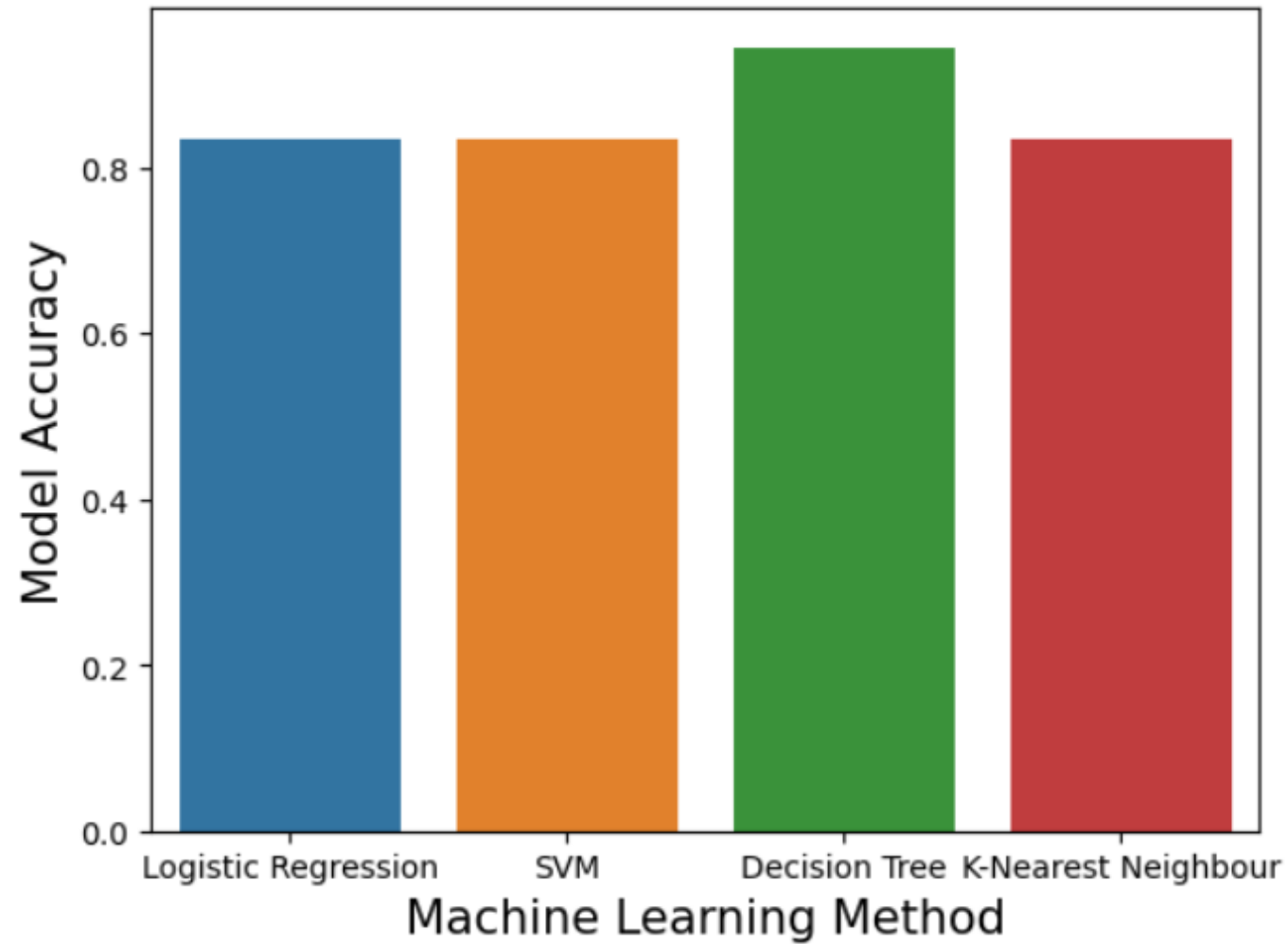


Payload Mass and Success Rate – All Sites

- **1 = Successful Mission Outcome 0 = Failed Mission Outcome**
- **Booster Version Analysis**
 - **B5** only one launch (3,500 Kg) and was successful
 - **V1.0** payload mass all < 2000 Kg and were unsuccessful
 - **V1.1** largest failure booster version across varied masses
 - **FT** and **B4** highest success rates with payload mass between 2000Kg and 6000Kg

Section 5

Predictive Analysis (Classification)

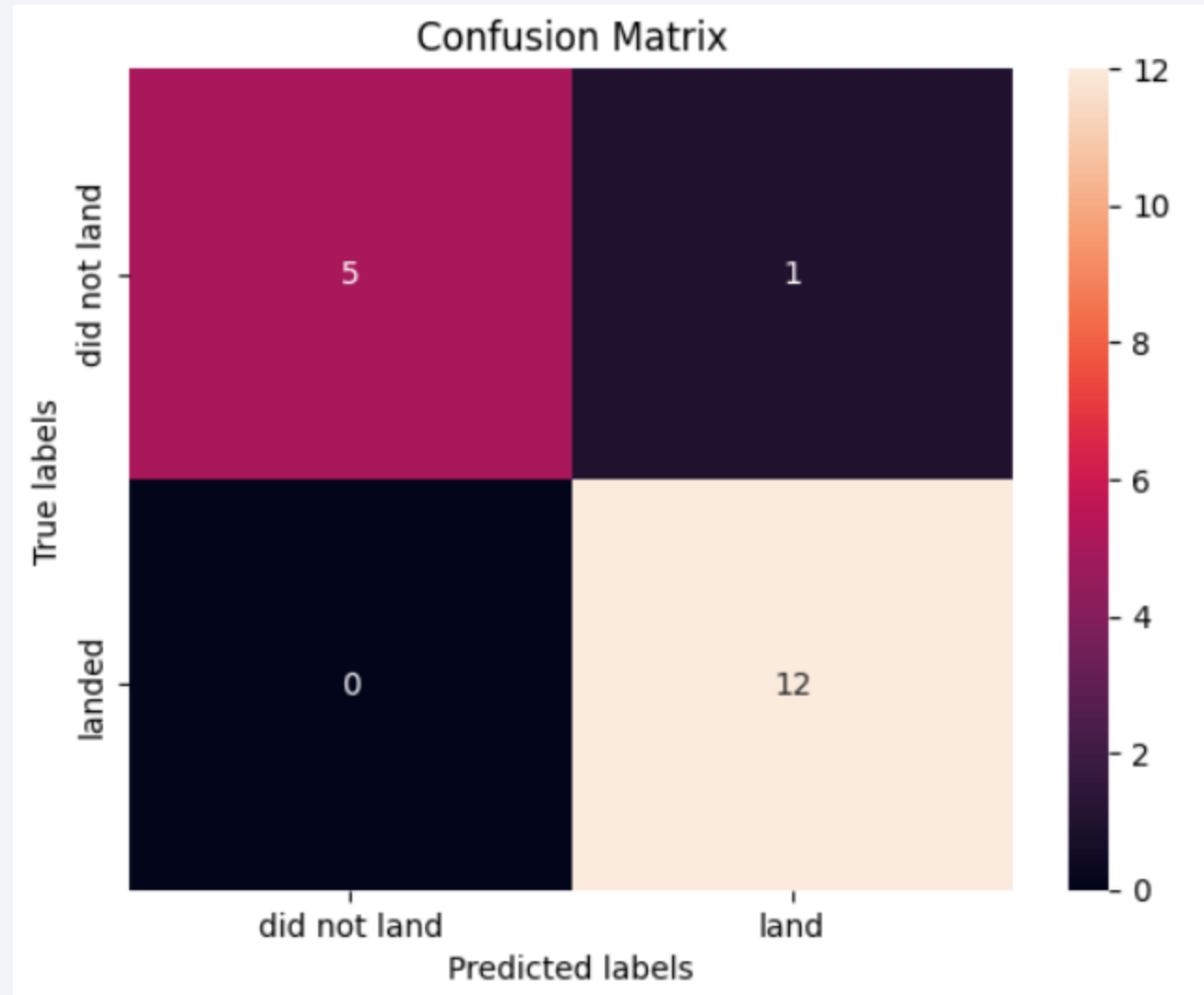


Classification Accuracy

- Small testing sample size (n=18)
- Most Models performed similarly with accuracy of 83.3% (15/18)
- Decision Tree performed best with accuracy of 94.4% (17/18)

Confusion Matrix

- Decision Tree performed the best
- **17 of 18 correctly predicted**
- Only **1 False Positive** and **0 False Negative** predictions
- Mitigating Type 1 Error is key
- **Accuracy = 0.944**
- **Precision = 0.923**
- **Recall = 1**
- **F1 Score = 0.96**



Conclusions

Discussion of Results:

- Falcon 9 launches have **improved over time**, increasing in average success rate from **0% to 80% over 2010-2020 time period**
- Launch sites **closer to equator** to take advantage of Earth rotational speed to save on fuel consumption
- **Close to coastline and away from cities + major transportation** to reduce collateral damage done to human-inhabited area
- **Heavier payloads** generally correlated with **higher success rate**
- **Very Low Earth Orbit (VLEO)** and **Sun Synchronous Orbit (SSO)** show **high success rates** across multiple launches
- **Decision Tree model** performed best, correctly labeling **17/18 outcomes** from testing set

Things to consider:

- **Small sample size** – more data should be collected to ensure relationships found remain consistent and models perform similarly
- **Multicollinearity** – features used to predict landing outcome show levels of correlation. For example, the orbit type for a specific launch seemed to affect the mass of payload used. This could adversely impact models that over-represent certain features in training
- **Potential for better models** – there is a possibility simpler models (Simple Linear Regression) could show similar results to models used, while improving explainability of results. On the flip side, more complex models (Neural Networks) may show better predicting power.

Thank you!

